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Quantification of latent travel demand across age groups with a focus on older adults: a theoretical framework with empirical evidence from China

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Highlights

- Operationalizes latent travel demand with a four-dimensional framework.
- Quantifies age-differentiated effects of health, functional, social and tech factors.
- Social participation dominates: 75.3% total and 86.6% active latent demand.
- Reveals distinct mechanisms: health barriers in middle-aged, functional capacity in 75+.
- Supports policies prioritizing social participation beyond infrastructure.

Abstract

Against the backdrop of global population aging, promoting mobility among older adults is essential for achieving active aging goals. Drawing on the latent travel demand perspective, this study develops and validates a four-dimensional framework—health barriers, functional capacity, social participation, and technology acceptance—to identify and quantify latent travel demand across age groups, with a focus on older adults. Age-stratified regression

analyses were conducted using 664 survey samples from Ningbo, China. The results show that social participation is the strongest and most stable predictor of latent travel demand across the 35–59, 60–74, and 75+ age groups. It contributes 75.3% of total latent travel demand and 86.6% of active travel latent demand, with each standard deviation increase associated with approximately 1.2–1.5 additional daily total trips and 0.6 additional active trips. The disaggregated analysis shows that social interaction is the strongest driving component of the Social Participation Index. In contrast, health barriers and functional capacity display distinct age-specific patterns: health barriers primarily reduce total trips in the 35–59 group by approximately 0.25 per standard deviation increase, whereas functional capacity exerts a stronger positive effect among adults aged 75+, adding approximately 0.6 daily trips per standard deviation improvement. Technology acceptance shows no significant effects across groups. The findings suggest that age-friendly policy should move beyond physical infrastructure improvement alone and place greater emphasis on promoting social participation. In Ningbo, policy priorities may include expanding senior universities and community activity centers, promoting neighborhood and community activities, developing flexible employment and volunteering opportunities, and providing assisted transportation services for adults aged 75 and above. Beyond the Ningbo case, the framework can be adapted to local contexts and integrated with end-point evaluation tools to support systematic assessment of age-friendly mobility policies.

Introduction

The world is experiencing rapid population aging. The global population aged 60 and over increased from 9% in 1995 to 13% in 2025 and is projected to exceed 16% by 2035 (United Nations, 2024). China's aging process is particularly notable in scale and speed. By the end of 2023, the population aged 60 and above in China reached 310 million, accounting for 22.0% of the total population (National Bureau of Statistics of China, 2024). The WHO framework emphasizes health, participation, and security, arguing that older adults should be active participants in social life rather than passive recipients of protection (WHO, 2002). China's policy similarly emphasizes both support and contribution in later life (Central Committee of the Communist Party of China & State Council, 2021), calling for a shift from passive protection toward proactive social participation. Ensuring adequate mobility among older populations therefore constitutes a fundamental prerequisite for achieving active aging objectives. However, surveys in China indicate that about 19% of older adults hardly go out weekly, fewer than half make three or more trips, and around 63% rely on public transportation for daily travel (Huashang News, 2025). Therefore, it is essential to study latent mobility demand among older adults in China and implement targeted measures to improve their mobility.

The value perception of older adult mobility has evolved from individual health maintenance to a driving force for socioeconomic development. Mobility has a direct impact on older adults' life functions, quality of life, and psychological health. Mobility loss among older adults is associated with substantially higher risks of dementia and coronary heart disease, with reported increases of up to 50% and 30%, respectively (Maresova et al., 2023). Higher mobility is associated with the ability of older adults to access medical services, participate in social activities, and maintain interpersonal relationships, forming an important foundation for preventing social isolation (Inoue et al., 1998; He & Heinen, 2026).

With the advancement of human rights discourse, older adult mobility has increasingly been recognized as a fundamental right warranting equal protection. In the mid-20th century, Nordic countries proposed the concept of normalization, emphasizing that policy-making should eliminate discriminatory treatment based on age, disability, and other factors (Tøssebro et al., 2012, Alousi-Jones et al., 2025). The United Nations 2030 Sustainable Development Agenda Goal 11 explicitly calls for providing safe, affordable, and accessible transport systems for all

people (United Nations, 2015). This rights-based value positioning highlights the special identity of older adults as a vulnerable group, often placing them alongside disabled groups in policy design considerations.

As the older adult population continues to expand, the strategic significance of mobility has increasingly manifested in the economic dimension, becoming a key driving force for the development of the silver economy. According to the European Commission, the silver economy captures the economic opportunities of population aging by transforming older adults' purchasing power into a source of growth (Commission, 2018). Data from Georgetown University McDonough School of Business (2018) indicates that the global silver economy market value reached 15 trillion dollars in 2020. The European silver economy is estimated at 3.7 trillion euros and is projected to contribute over 5.7 trillion euros to the European economy by 2025. Within this expanding market, mobility emerges as a critical enabler that promotes the economic participation of older adults in consumption, tourism, and employment (Maresova et al., 2023).

Beyond economic benefits, improving older adult mobility can also produce significant cost control effects. From a social perspective, enhancing mobility can reduce medical and long-term care costs, easing pressure on healthcare systems (De Nardi et al., 2016). Japanese researchers found that lower travel frequency is associated with higher care costs, suggesting that public transport's health benefits should be included in policy evaluation (Hirai, 2019).

Given the multiple values of older adult mobility, governments worldwide have adopted policies to address mobility challenges among older populations. Japan has constructed a systematic transportation age-friendly system since the implementation of the Barrier-Free Act in 2006 (Nippon.com, 2016). The European Union has established accessibility standards covering products and services through the European Accessibility Act (European Disability Forum, 2023). The United States requires public transportation vehicles to have accessibility features under the Americans with Disabilities Act (National Aging and Disability Transportation Center, 2024). China has promoted the Barrier-Free Environment Construction Law and the 2024 work plan to expand and improve barrier-free transportation services (Ministry of Transport of China, 2024).

Transportation support measures for older adults can be divided into four categories. The first category includes functional compensation measures for older adults with physical limitations, such as wheelchair access and low-floor buses in barrier-free transportation systems (Ravensbergen et al., 2022, Yin et al., 2026). The second category involves age-friendly measures that improve comfort and safety for older adults through extended crossing times, rest seating, and large-font signage (Luiu et al., 2016, Delbosc and Currie, 2011). The third category includes active aging promotion measures that encourage older adults to participate in social activities and enhance their sense of value and responsibility through tourism charter services and employment transportation support (Wong et al., 2018). The fourth category involves technology innovation measures that enhance older adults' travel capabilities through smart wearables, autonomous driving, and AI-based recommendation systems (Fagnant and Kockelman, 2015, Harper et al., 2016, Liu et al., 2019, Oh et al., 2026).

The first two categories have been more extensively institutionalized, whereas the third and fourth remain underdeveloped and less systematically implemented (Lee & Lee, 2025). This disparity partly reflects a policy tendency to treat older adults as a homogeneous vulnerable group requiring protection. Existing policies often overlook heterogeneity in health status, functional capacity, social participation, and technology acceptance, although many older adults still maintain strong travel capabilities. Such simplified policy orientations may lead to inefficient resource allocation and suppress latent travel demand. Moreover, current policy frameworks lack empirical evidence on which interventions should receive priority investment, a gap that theoretical research has yet to address.

The policy gaps identified above reflect underlying theoretical limitations. Several frameworks have been developed to understand older adult mobility. Social exclusion theory emphasizes that mobility constraints restrict social participation, and Lucas (2012) established the link between transport disadvantage and social exclusion. The capability approach focuses on individuals' ability to achieve functional freedom in travel (Sen, 1999). Based on this approach, Li (2024) proposed a framework from transport equity to accessibility capability for older adult transportation planning. The transport disadvantage theory examines the travel difficulties of vulnerable groups (Broome et al., 2009) and suggests that transport inclusivity can improve older adults' travel opportunities through opportunity accessibility, physical accessibility, age-friendly informatization, and affordability.

Despite these contributions, existing research has two core limitations. First, it remains conceptually ambiguous in distinguishing older adult mobility from disability mobility. Many policies rely on administrative categories rather than functional assessment, which overlooks heterogeneity in mobility impairments and travel capabilities among older adults. This may hinder the shift from viewing older adults as passive care recipients to recognizing them as active social participants. Existing studies also focus mainly on developed countries, while evidence from low- and middle-income contexts remains limited.

Second, existing frameworks do not adequately quantify the relationship between latent travel demand and policy intervention effects. Most studies provide qualitative assessments or limited empirical analysis, but lack predictive mechanisms for converting unmet mobility needs into measurable travel outcomes. This is important because older adults may express a desire to travel while still avoiding actual trips (Jahangir et al., 2024). Without predictive modeling, policymakers cannot easily assess intervention effectiveness or allocate resources efficiently. End-point evaluation tools are already available. For example, the WHO Health Economic Assessment Tool (HEAT) estimates health benefits, air pollution reduction, and carbon emission reduction based on changes in walking and cycling distances (Kahlmeier et al., 2017). However, HEAT treats behavioral change as an input and cannot explain how policy interventions generate such change. End-point tools can estimate the benefits of increased travel, but they cannot identify which interventions are most likely to increase travel.

Among existing theoretical contributions, the latent travel demand framework proposed by Miyoshi and Nitta (1995) is particularly useful for bridging theory and policy. It distinguishes four types of latent demand, including demand suppressed by physical disabilities, age-related functional decline, social role transitions such as retirement, and emerging technologies. This typology corresponds to the four categories of policy measures and provides a coherent lens for linking interventions with travel demand. However, the framework remains mainly descriptive. It lacks measurable indicators and quantification methods for assessing the relative importance of each constraint dimension and supporting policy priority setting.

To address these gaps, this study operationalizes the framework by transforming it into a practical policy analysis tool. Specifically, it pursues three objectives: 1) to construct a four-dimensional indicator system consisting of the Health Barrier Index, Functional Capacity Index, Social Participation Index, and Technology Acceptance Index; 2) to quantify the relative influence of each constraint dimension on travel behavior through age-stratified regression analyses using survey data from Ningbo, China. The analysis covers the 35 to 59, 60 to 74, and 75 and above age groups, with a focus on older adults; 3) to provide evidence-based policy recommendations for enhancing mobility and social participation among older adults in Ningbo. The contributions are threefold. First, it transforms a classical theoretical framework into measurable and quantifiable indices. Second, it provides empirical evidence for policy priority setting. Third, it establishes an upstream analytical framework that can be integrated with end-point evaluation tools such as the WHO HEAT, forming a complete assessment chain from policy intervention to behavioral response to health and environmental benefits.

Section snippets

Conceptual framework of latent travel demand

Based on the study by Miyoshi and Nitta (1995), the latent travel demand of older adults can be elucidated through the conceptual framework illustrated in Fig. 1. It includes three age cohorts, namely young adults, middle-aged individuals, and older adults, as well as five curves representing mobility levels under different circumstances. The curves serve as illustrative trajectories reflecting how various influencing factors affect mobility demand, rather than precise calibrations based on ...

Data collection

The empirical analysis uses specialized survey data collected in Ningbo, a sub-provincial city in the Yangtze River Delta Economic Zone on China's eastern coast (Fig. 2). Ningbo provides an appropriate empirical context because it combines pronounced population aging with strong economic vitality. Residents aged 60 and above account for 30.1% of the population, exceeding the national average of 22%, while per capita disposable income is over 50% higher than the national average (National Bureau ...

Impact of mobility constraint indices on travel behavior

Table 3 presents unadjusted descriptive comparisons of mean daily trip frequency across categorized mobility constraint index groups and travel modes. Because these comparisons do not control for covariates, they should be interpreted as descriptive group differences rather than independent effects of the indices. Daily travel frequency was calculated as the reported number of trips during the survey week divided by seven. Sample sizes, reference groups, and index-specific thresholds are shown ...

Theoretical and practical contributions

This study extends the conceptual framework of latent travel demand among older adults proposed by Miyoshi and Nitta (1995) by translating it into an operational empirical approach. Although the original framework identified different forms of latent demand and related countermeasures, it did not provide a measurable indicator system or a quantitative method for policy analysis. This study addresses this gap by constructing four composite indices—the Health Barrier Index (HBI), Functional ...

Conclusions

Against the backdrop of global population aging, promoting mobility among older adults is essential to active aging. This study translates the latent travel demand framework into an empirical policy analysis tool by constructing four composite indices—the Health Barrier Index (HBI), Functional Capacity Index (FCI), Social Participation Index (SPI), and Technology Acceptance Index (TAI)—and applying age-stratified regression models to survey data from 664 respondents in Ningbo. The analysis ...

CRedit authorship contribution statement

Yanhong Yin: Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Formal analysis, Data curation, Writing – review & editing. **Yang Xi:** Validation, Methodology, Funding acquisition,

Conceptualization. ...

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